

B2 R2 Responses Master

Paper: Governance Concentration Beyond Token Allocation: An Institutional Design Analysis of DePIN and DeFi **Author:** Zach Zukowski (Tokenization Systems; zach@tokenization.systems) **Journal:** Frontiers in Blockchain **Round:** R2 (response to Reviewer 1 second-round feedback) **Date:** 2026-05-17 (final; Phase 0 data collection complete)

Preamble

We thank both reviewers for their continued engagement with the manuscript. Reviewer 2 endorsed publication in R2. The revisions below address Reviewer 1's four remaining issues at the level of granularity raised. The R2 cycle treats Reviewer 1's specific findings as one class (manuscript-vs-data-of-record drift) and resolves them with a comprehensive reproduce-and-sync pass per the reviewer's general suggestion. Beyond Reviewer 1's specific issues, the R2 cycle also strengthens the philosophical-methodological framework substantially (Section 2.9.6 inter-lens relationships; Section 2.10.1 two-layer framing; Section 4.5 comparative methodology; 21 new citations across philosophy, mechanism design, voting theory, and institutional economics; falsifiable forward claims; data-availability statement; multiple-comparisons correction note).

The most consequential single result of the R1 to R2 reproduce-and-sync is a substantive interpretive change in Section 3.5: with corrected holding HHI baselines for Uniswap and Optimism, both protocols flip from delegation-mediated dispersion to delegation amplification. The post-2026-05-17 deep-PCA-audit cycle (commit 3a8c3065; 2026-05-19) further expanded Table 7 from N=8 to N=10 (added GMX and ENS via Tally additions per the 12-month governance-activity gate). The R2-2026-05-21 voting-HHI methodology refresh (per EC-2026-05-21-B2-Voting-HHI-Sampling-Methodology-Floor) expanded Snapshot pulls to ALL proposals in the 12-month rolling window 2025-05-22 to 2026-05-22 and Tally pulls to top-1000 delegates per protocol; the refresh surfaced GMX as a 2nd structural-exception counterexample (FLIP from 1.2x amplification at top-100 sample to 0.87x dispersion at top-1000 sample; sampling-depth artifact revealed). **The 2026-05-22 multi-source PCA-symmetric extension cycle (commit 1751fd5c; Cycle 15 below) further expanded Table 7 from N=10 to N=13 by adding three Solana-native protocols (JUP delegated-vote 0.057x dispersion; HNT VSR lockup-weighted 0.35x-0.53x sensitivity-range dispersion; DRIFT VSR lockup-weighted 1.6x amplification).** The R2 manuscript currently documents a **predominant-amplification finding with four structural exceptions** across nine of thirteen protocols in the Cycle-15 Table 7 sample (amplifier range 1.6x DRIFT to 9.9x Compound; mean approximately 4.9x), with ENS at 0.45x (Section 3.5.1 mature-delegate-program), GMX at 0.87x (Section 3.5.1.1 methodology-sensitive), HNT at 0.35x-0.53x (Section 3.5.4 Solana VSR-lockup-weighted), and JUP at 0.057x (Section 3.5.4 Solana delegated-vote + broad-airdrop-user-base; the most-extreme dispersion outlier in the cross-section, 8x more extreme than ENS) as the four structural exceptions. A robustness check under symmetric PCA-exclusion methodology supports the finding holding across all Tally-sourced protocols even when foundation and aggregation-contract delegates are excluded from voting HHI (Section 3.7), extended in Cycle 15 across three governance-data

sources: Tally all-delegates depth (AAVE N=150,911; COMP N=18,627; UNI N=48,707; ARB N=437,453; all four sub-0.005 HHI shift vs cycle-13 top-1000 baseline); Snapshot signer-side functional-PCA exclusion via new sibling canonical store `exclusions_log_signer.csv` (4 confirmed entries: Compound Foundation; Balancer DAO multisig; Kevin Owicki Bitcoin Maintainer; WXM Hedgey-vested insider); and Solana on-chain governance program parsing (JUP via Dune `jupiter_solana.govern_call_setvote`; HNT and DRIFT via Helius RPC parsing of SPL Governance VoteRecordV2 accounts). Section 3.5.1 documents the ENS case as institutional-design-tractable existence-proof; Section 3.5.1.1 documents the GMX case as methodology-sensitive counterexample; NEW Section 3.5.4 documents the within-Solana governance concentration heterogeneity (VSR-vs-delegated weight design as primary axis); NEW Section 3.5.4.1 documents the Drift Foundation proposal-authorship-vs-vote-weight decoupling pattern (single Foundation submitter authored 11 of 13 DRIFT Improvement Proposals while holding only 0.18 percent of total vote weight); NEW Section 3.5.5 documents the cross-protocol institutional-governance-investor pattern (PGov + Tane Governance + Arana Digital approximately 17 percent combined Snapshot voting power across 6 to 7 DAOs; candidate Fifth Concentration Axis). The four structural-exception cases preserve the dispersion pattern as informative counterexamples rather than refuting the broader predominant-amplification thesis.

Issue 1: Table 7 holding-HHI inconsistency (interpretive flip)

Reviewer comment (verbatim, key points)

Table 7 still uses the old holding-HHI baselines for some protocols, even though the corresponding HHI values have been revised elsewhere in the paper and repository. ... Using the revised holding-HHI value would make the voting/holding ratio roughly 2.7X rather than 0.84X [for Uniswap]. Optimism's Table 4/processed-data holding HHI is approximately 0.009, while Table 7 still uses 0.042. ... Lido's ratio also appears lower when the revised holding HHI is used. ... I recommend that Table 7 be fully recomputed using the same proper holding-HHI values used in Table 4 and in the regression dataset.

Response

The reviewer is correct on all three flagged cases. Table 7 has been fully recomputed using post-exclusion holding HHIs consistent with Table 4. The Compound and Arbitrum rows continue to use Snapshot-sourced voting HHIs per the R1 author rationale (larger active-voter pools than Tally's top-100 sampling); voting HHI source assignments are unchanged from R1. Holding HHIs are now uniformly post-exclusion across Table 7.

Table 7 before/after (4-cycle cascade through Cycle 15):

Pr ot oc ol	S e c t o r	R1 Ho ldi ng	R2- 2026- 05-17 Holdin g	R2-2026-05- 21 Holding (cycle-13)	R2-2026-05- 22 Holding (Cycle-15)	Vo tin g H I		R 1 R 2026- 05-17 Ratio	R2- 2026- 05-21 Ratio	R2- curre nt Ratio	Di re cti on
						(c urr ent)	R at io				
DI M O	D e P I N	0.0 38	0.038	0.025	0.025	0.2 28	6. 0 x	6.0x	9.1x	9.1x	a m pli fie s
Li do	D e F i	0.0 18	0.013	0.008	0.008	0.0 50	4. 8 x	6.8x	11.4x → 6.3x	6.3x	a m pli fie s (c yc le- 13 re fr es h sh ift ed fr o m 11 .4 x to 6. 3x at Sn ap sh ot N =3

Protocol	Sector	R1 Holding	R2-2026-05-17 Holding	R2-2026-05-21 Holding (cycle-13)	R2-2026-05-22 Holding (Cycle-15)	Voting Rights		R2-2026-05-17 Ratio	R2-2026-05-21 Ratio	R2-current Ratio	Direction
						HI	R1				
Wealthier X-M	DPEIN	0.148	0.148	0.148	0.148	0.556	3.3x	3.8x	3.8x	3.8x	amplifies (cycle-13 refresh: voting 0.486 to 0.556 at Snapshot N=73)
Compo	DE F	0.028	0.009	0.009	0.009	0.089	1.9x	5.7x → 9.9x	5.7x → 9.9x	9.9x	amplifies

Pr ot oc ol	S e c t o r	R1 Ho ldi ng	R2- 2026- 05-17 Holdin g	R2-2026-05- 21 Holding (cycle-13)	R2-2026-05- 22 Holding (Cycle-15)	Vo tin g H I (c urr ent)	R 1 R at io	R2- 2026- 05-17 Ratio	R2- 2026- 05-21 Ratio	R2- curre nt Ratio	Di re cti on
un d	i										s (c yc le- 13 re fr es h: vo tin g 0. 05 3 to 0. 08 9 at Sn ap sh ot N =1 38)
A av e	D e F i	0.0 20	0.020	0.013	0.013	0.0 58	3. 8 x	3.8x	5.9x → 4.8x	4.4x	a m pli fie s (C yc le- 15

						Voting HI	R1	R2-2026-05-17 Ratio	R2-2026-05-21 Ratio	R2-current Ratio	Direction
Protocol	Score	R1 Holdi ng	R2-2026-05-17 Holdi ng	R2-2026-05-21 Holding (cycle-13)	R2-2026-05-22 Holding (Cycle-15)	(current)					:
											vo
											tin
											g
											0.
											06
											2
											cy
											cl
											e-
										13	
										to	
										p-	
										10	
										00	
										to	
										0.	
										05	
										8	
										all	
										-	
										de	
										le	
										ga	
										te	
										s	
										N	
										=1	
										50	
										,9	
										11	
										)	
Units wap	Defi	0.032	0.010	0.010	0.010	0.027	0.84x	2.7x	2.7x	2.7x	ampli fies

						Voting Right	R1 Ratio	R2-2026-05-17 Ratio	R2-2026-05-21 Ratio	R2-current Ratio	Direction
Product	Scenario	R1 Holding	R2-2026-05-17 Holding	R2-2026-05-21 Holding (cycle-13)	R2-2026-05-22 Holding (Cycle-15)	(current)					
Arbitrum	Infra	0.012	0.012	0.012	0.038	4.3x	4.4x → 3.1x	3.1x			(FLOP at R2-2026-05-17) amplified (cycle-13 refresh: voting 0.052 to 0.038 at

	S	R1	R2-			V	R				
Pr ot oc ol	e c o r	H o l d i n g	2026- 05-17 Holdin g	R2-2026-05- 21 Holding (cycle-13)	R2-2026-05- 22 Holding (Cycle-15)	o t i n g H I (c urr ent)	1 R at io	R2- 2026- 05-17 Ratio	R2- 2026- 05-21 Ratio	R2- curre nt Ratio	Di re cti on
O pti mi s m	I n f r a	0.0 42	0.009	0.009	0.009	0.0 33	0. 7 9 x	3.7x	3.6x	3.6x	Sn ap shot N=5, 466) a m pli fie s (F LI P at R 2- 20 26 - 05 - 17)
G M X	D e F i	n/a	n/a	0.065 (Tally May 2026)	0.065	0.0 57	n/ a	n/a	0.87x (FLIP)	0.87x	DI S P E R S E S (c yc

Table 1: Performance Metrics and Analysis											
Project ID	System Configuration			Performance Data (Q1-Q2)		Resource Utilization		Efficiency Ratios		Compliance & Audit	
	Processor	Memory	Storage	Q1 Avg	Q2 Avg	CPU %	Mem %	Q1:Q2	Q1:Q2	Score	Notes
P001	Intel Xeon	64GB	SSD 1TB	1200	1150	45%	60%	1.05x	0.95x	92	Minor config change
P002	AMD Ryzen	32GB	HDD 5TB	950	980	30%	40%	0.98x	1.02x	88	Stable performance
P003	ARM Cortex	16GB	SSD 512GB	800	820	20%	35%	1.02x	0.98x	85	Optimized for low power
P004	Intel Xeon	128GB	SSD 2TB	1500	1450	55%	70%	1.03x	0.97x	95	High performance tier
P005	AMD Ryzen	64GB	SSD 1TB	1100	1080	40%	55%	1.01x	0.99x	90	Consistent results
P006	ARM Cortex	32GB	SSD 256GB	750	760	15%	30%	1.00x	1.00x	80	Edge computing optimized
P007	Intel Xeon	96GB	SSD 1.5TB	1350	1300	50%	65%	1.04x	0.96x	93	Advanced analytics workload
P008	AMD Ryzen	48GB	SSD 768GB	1000	990	35%	45%	1.01x	0.99x	89	Medium scale deployment
P009	ARM Cortex	24GB	SSD 128GB	700	710	18%	28%	1.02x	0.98x	78	IoT sensor network node
P010	Intel Xeon	192GB	SSD 3TB	1600	1550	60%	75%	1.05x	0.95x	97	Enterprise data center

Protocol	Security	R1 Holding	R2-2026-05-17 Holding	R2-2026-05-21 Holding (cycle-13)	R2-2026-05-22 Holding (Cycle-15)	Voting History (current)	R1 Ratio	R2-2026-05-17 Ratio	R2-2026-05-21 Ratio	R2-current Ratio	Direction
ogram counter example)											
JUP	DEFI	n/a	n/a	n/a	0.096 (Cycle-15 Solana addition; Helius DAS API)	0.0055 (Solana delegated - vote per jupiter_solana.govern_	n/a	n/a	n/a	0.057x	DISES (most extreme measure dispersion outlier in

Pr ot oc ol	S e c t o r	R1 Ho ldi ng	R2- 2026- 05-17 Holdin g	R2-2026-05- 21 Holding (cycle-13)	R2-2026-05- 22 Holding (Cycle-15)	Vot ing H I (c urr ent)					R 1 R at io	R2- 2026- 05-17 Ratio	R2- 2026- 05-21 Ratio	R2- curre nt Ratio	Di re cti on
						call _s et vo te)									
H N T	D e P I N	n/ a	n/a	n/a	0.075 (Cycle-15 Solana addition)	0.0 26 - 0.0 39 (V S R loc ku	n/ a	n/a	n/a	n/a	0.35x -0.53 x	D IS P E R S E S (S ol			

Pr ot oc ol	S e c t o r	R1 Ho ldi ng	R2- 2026- 05-17 Holdin g	R2-2026-05- 21 Holding (cycle-13)	R2-2026-05- 22 Holding (Cycle-15)	Vo tin g H I (c urr ent)	R 1 R at io	R2- 2026- 05-17 Ratio	R2- 2026- 05-21 Ratio	R2- curre nt Ratio	Di re cti on
						p- we ig ht vi a He liu s R P C; 14. 8 % clo se d- po sit io n se nsi tiv ity ra ng e)					an a V S R- lo ck u p- w ei gh te d; Se cti on 3. 5. 4)
D R IF T	D e f i	n/ a	n/a	n/a	0.053 (Cycle-15 Solana addition)	0.0 83 (V S R- we ig ht	n/ a	n/a	n/a	1.6x	a m pl ifi es (N E W

Protocol	Snapshot	R1 Holding	R2-2026-05-17 Holding	R2-2026-05-21 Holding (cycle-13)	R2-2026-05-22 Holding (Cycle-15)	Voting HHI (current)	R1 Ratio	R2-2026-05-17 Ratio	R2-2026-05-21 Ratio	R2-current Ratio	Direction
						ed					low
						vi					wes
						a					t
						He					a
						liu					m
						s					pl
						R					ifi
						P					er
						C					;
						Vo					So
						te					la
						Re					na
						co					V
						rd					S
						V2					R-
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Note: Tally rows refreshed at cycle 13 to top-1000 delegate samples (Aave further refreshed at Cycle 15 to all-delegates N=150,911); voting HHIs and ratios reflect post-refresh values. Snapshot rows refreshed at cycle 13 to all-proposals-in-12-month-rolling-window methodology; per-protocol voting-HHI shifts documented in EC-2026-05-21-B2-Voting-HHI-Sampling-Methodology-Floor. Solana rows added in Cycle 15 (commit 1751fd5c) via on-chain

governance program parsing (JUP via Dune decoded namespace; HNT and DRIFT via Helius RPC SPL Governance VoteRecordV2 accounts). The Aethir Table 4 row was also updated in Cycle 15 Pass 6 audit from HHI 0.0868 to 0.0948 reflecting the post-cycle-4 0xfc78 Safe multisig exclusion that the caption already documented; Table 4 row data now consistent with caption + canonical replication CSV.

Substantive interpretive consequence. The flip is not a typo fix: it changes the Section 3.5 finding from “delegation amplifies unevenly, with two infrastructure protocols showing opposite patterns” (the R1-era framing) to “delegation amplifies governance concentration in nine of ten protocols sampled, with one structural exception (ENS at 0.57x dispersion) and with magnitude varying by institutional design within sector” (the R2-current near-universal-amplification framing, post deep-PCA-audit cycle 2026-05-19 sample expansion). The R2-2026-05-17 framing (“universal-amplification across all eight protocols”) was an intermediate-state framing valid at this letter’s authoring date; the post-2026-05-17 deep-PCA-audit expanded the sample to N=10 and surfaced the ENS structural counterexample, reframing the thesis to near-universal. We have rewritten Section 3.5 (opening paragraph, DeFi paragraph, Optimism/Arbitrum paragraph, Citizens’ House/Dual Governance paragraph, Fritsch et al. extension paragraph), Section 3.5.1 ENS counterexample subsection (added in deep-PCA-audit cycle), Section 3.5.2 ve-token class subsection, abstract finding 4, Section 1.4 finding 4, and the Section 4.1 design-hypotheses synthesis to reflect the near-universal framing.

On Lido. The reviewer’s observation that Lido’s ratio “appears lower” with the revised holding HHI is opposite to what our data produces. The R1-corrected Lido holding HHI (Table 4 value 0.013 at this letter’s authoring date) yielded an amplification ratio of 6.8x. The R2-2026-05-17 Table 7 reported 6.8x. The post-2026-05-17 deep-PCA-audit cycle (commit 3a8c3065; 2026-05-19) further refined Lido holding HHI to 0.008 (per PCA exclusion of additional Lido DAO addresses identified in the 126-address audit). The cycle-13 voting-HHI methodology refresh (commit 99036422; 2026-05-21) expanded the Lido Snapshot pull from sample-bounded (N=262 across 5 proposals) to comprehensive (N=333 across all proposals in the 12-month rolling window), yielding voting HHI 0.050 and amplification ratio 6.3x. The R2-2026-05-17 6.8x and the post-deep-PCA-audit 11.4x are preserved as audit-trail in the F-B2-9 cascade chain documented in docs/KEY_FINDINGS.md; the R2-current 6.3x at cycle-13 sampling depth supersedes per EC-2026-05-21-B2-Voting-HHI-Sampling-Methodology-Floor. The prior 6.8x figure is preserved as audit-trail in the F-B2-9 cascade chain documented in docs/KEY_FINDINGS.md. If the reviewer is computing a different Lido holding HHI from an alternative exclusion list (for example, one that does not exclude the Lido DAO Aragon treasury), we would welcome specification of the methodology; the canonical exclusion set we apply is documented in Supplementary File S6.

On the Uniswap counterexample. The pre-correction Uniswap result (0.84x) had been cited as evidence that Uniswap’s mature delegate program disperses governance power. With the burn address exclusion applied (the canonical 0x000...000dead holding 102.5M UNI, 11.3 percent of supply, structurally unable to participate in governance), Uniswap joins the universal-amplification pattern at 2.7x. We document this transition explicitly in Section 3.5 because the prior result was load-bearing in the dispersion-via-mature-delegate-program literature; the corrected analysis indicates the prior result was driven by including a wallet that structurally cannot vote in the holding denominator.

Compound Foundation as PCA at the voting layer. Phase 0 fact-checking of the Tally API surfaced a substantive methodological finding: the top COMP Tally delegate (21.5% of total delegated voting power in the May 2026 pull) is Compound Foundation itself, structurally a protocol-controlled entity. Applying PCA exclusion symmetrically at the voting layer (consistent with the holding-side methodology) drops Compound's voting HHI from 0.078 to 0.052 and the amplification ratio from 1.89x to 1.85x. The R1 Snapshot-based COMP voting HHI value (0.053) is essentially equivalent to the PCA-excluded Tally value (0.052); the R1 methodology choice already approximated the PCA-exclusion outcome. We document this analysis as a Section 3.7 PCA-symmetric robustness paragraph showing universal amplification holds across all 5 Tally-sourced protocols (Compound 1.85x, Aave 2.26x, Uniswap 2.72x, Optimism 4.06x, Arbitrum 3.01x) even under symmetric PCA exclusion.

Manuscript changes: Table 7 rows updated (UNI, OP, LDO holdings + ratios); Table 7 caption updated with explicit post-exclusion methodology note; Section 3.5 paragraphs 1-5 rewritten; abstract finding 4 rewritten; Section 1.4 finding 4 rewritten; Section 3.7 PCA-symmetric robustness paragraph added; Section 2.10.3 Tally data drift methodology note added (documents the March 2026 to May 2026 delegate-pool drift; the R2 manuscript uses the March 2026 snapshot consistent with the rest of the dataset, with May 2026 results reported as supplementary robustness check); Figure 6 regeneration pending Phase 4 figure-regeneration cycle.

Issue 2: Curve Top-10 column + CRV text disambiguation

Reviewer comment (verbatim)

In Table 4, Curve's HHI has been corrected to the post-exclusion raw-CRV value, but its Top-10 value is still the pre-exclusion one. The repository reports a substantially lower post-exclusion Top-10 value for Curve than the value shown in the table. In several places in the text, CRV is reported as HHI = 0.171, even though Table 4 now reports raw CRV, with HHI = 0.017. If this is supposed to be about veCRV, please mention it explicitly in the text.

Response

The reviewer is correct on both points. Table 4 Curve Top-10% has been corrected from 60.7% (pre-Convex-exclusion) to 34.4% (post-Convex-exclusion, matching the regression dataset). The CRV-vs-veCRV disambiguation has been implemented across all four textual surfaces, with the 0.171 proxy replaced by a directly-computed veCRV voting concentration value (0.26) per Phase 0 Tier C1 analysis below.

Direct veCRV voting concentration computation (Phase 0 Tier C1). The R1 manuscript used 0.171 as a proxy for veCRV-weighted voting concentration. The 0.171 figure is the raw CRV holding HHI INCLUDING the Convex veCRV custody position (which holds substantial CRV that has been locked into veCRV). For R2 we compute the actual veCRV-weighted voting concentration directly via Dune analysis of the Curve VotingEscrow Deposit events at 0x5f3b5dfb7b28cbbd7faba78963ee202a494e2a2. Aggregating cumulative CRV deposits across the top-50 historical veCRV lockers yields HHI = 0.26 (Convex CRV Locker: 48.0% of top-50 deposit-weighted share; Yearn yveCRV: 13.5%; Stake DAO sdCRV: 9.5%; the remaining 47

lockers each less than 4%). The 0.26 direct computation captures the actual gauge-weight voting concentration that veCRV produces and is substantively higher than the 0.171 proxy.

Disambiguation strategy applied. We have made the raw-vs-veCRV distinction explicit at every textual surface where CRV is cited:

- **Table 4 (row):** HHI = 0.017 (raw post-exclusion); Top-10% = 34.4% (post-exclusion).
- **Table 4 (footnote methodology notes):** HHI for Curve reflects raw CRV holdings after excluding the Convex veCRV custody position; veCRV-weighted voting concentration is approximately 0.26 computed from cumulative CRV deposit volume across the top-50 historical veCRV lockers; methodology discussed in Section 4.1.
- **Section 3.2 (DeFi range description):** DeFi HHI ranges from 0.005 (Hyperliquid) to 0.117 (Jupiter); CRV's raw 0.017 places it mid-range. The 0.26 veCRV voting concentration is referenced with explicit methodology cite.
- **Section 4.1 (publicity-territory list):** CRV removed from the upper-end-of-distribution territory list (which is now Livepeer 0.199, WeatherXM 0.148, Aethir 0.153). The veCRV discussion is reserved for the Section 4.1 ve-locking paragraph.
- **Section 4.1 (Constitutional Transparency / Pettit Non-Domination ve-locking discussion):** Both surfaces distinguish raw post-exclusion CRV holding HHI (0.017) from veCRV-weighted voting concentration (0.26), with the gap attributed to lock-duration multipliers + Convex meta-governance aggregation.

The Aethir HHI in Table 4 was also corrected: the R1 footnote value of 0.171 reflected the March 2026 cascade snapshot; the R2 direct Dune pull (May 2026) yields 0.153. Both reflect raw top-1000 holder concentration; the drift between snapshots is documented in the Section 2.10.3 snapshot-discipline note.

Manuscript changes: Table 4 CRV Top-10% cell (60.7% to 34.4%); Section 3.2 DeFi-range paragraph rewritten with disambiguation; Section 4.1 publicity territory list updated; Section 4.1 ve-locking paragraph and Pettit non-domination paragraph updated to use 0.26 (direct computation); Table 4 methodology notes updated with explicit veCRV decomposition (Convex 48%, Yearn 14%, Stake DAO 9%); supplementary exhibit veCRV_voting_concentration_2026-05-17.csv ships with R2 submission.

Issue 3: Table 5 N=40 and TT-expanded subsidy reproducibility

Reviewer comment (verbatim)

First, the HHI-Gini row reports N = 40, but the current data seem to contain missing Gini values for some protocols. Second, the TT-expanded subsidy analysis is still not straightforwardly reproducible from the repository files. The CSV contains fewer observations than the N reported in the manuscript. In addition, the conclusions drawn appear to be based on the old numbers.

Response

The reviewer is correct on both sub-issues. We have addressed both in the R2 cycle.

HHI-Gini row N=40 issue. Three DePIN protocols (Aethir, Hivemapper, io.net) appeared in the R1 Table 4 footnote with HHI values but were noted as lacking the holder-level Gini and percentile data required for Table 4 reporting. The R1 cycle inconsistently reported N=40 for the HHI-Gini correlation despite this gap. For R2:

- Hivemapper (HONEY; Solana SPL): pulled via Helius DAS API (90,680 unique holders). HHI 0.017 (matches cascade methodology), Gini 0.92, Top-1% 6.7%, Top-10% 33.7%. Added as full Table 4 row.
- io.net (IO; Solana SPL): pulled via Helius DAS API (84,861 unique holders). HHI 0.111 (matches cascade exactly), Gini 0.95, Top-1% 30.0%, Top-10% 63.7%. Added as full Table 4 row.
- Aethir (ATH; Ethereum ERC-20): pulled via Dune erc20_ethereum.evt_transfer net-balance aggregation (top-1000 holders). HHI 0.153, Gini 0.98, Top-1% 33.6%, Top-10% 83.8%. Added as full Table 4 row.

All three protocols now appear as full Table 4 rows with complete holder-level data; the R2 manuscript reports HHI-Gini correlation at N=40 with both metrics measured for every protocol. The R1 footnote noting their lack of Gini/percentile data has been replaced with a footnote documenting the R2 supplementary holder-list collection.

TT-expanded subsidy CSV reproducibility. The reviewer correctly identifies that the R1 manuscript's reported N=22 for the TT-expanded subsidy test does not match the row count of subsidy-ratio-populated observations in `tokenterminal_financials.csv`. The R2 audit (Phase 0 Tier A2) confirmed that the canonical regression dataset contains 19 protocols with non-null Token Terminal subsidy_ratio data at this letter's authoring date; the manuscript's N=22 was a stale count from an earlier draft. The post-2026-05-17 cycle 6 finalization workstream (cycle 12 net-flow operationalization 2026-05-19) further expanded the sample to N=20 with Pearson $r=0.12$, $p=0.61$. For R2-current:

- The Table 5 "Subsidy ratio (TT expanded)" row is updated to N=20 with corresponding Pearson $r=0.12$, $p=0.61$ (computed directly from the 20 protocols with TT subsidy data in the post-2026-05-17 canonical regression dataset). Closely matches the manuscript's previously-reported $r=0.095/0.10$ line, confirming the qualitative conclusion (null cross-sector association) but with the corrected N.
- The Section 3.7 paragraph documenting the TT-expanded robustness check is updated to N=20 with $r=0.12$.
- A new supplementary sensitivity check is added: combining TT subsidy with on-chain subsidy data for protocols where TT data is not available yields N=26 across both methodologies (post-2026-05-17 cycle 6 expansion from N=25; cycle 12 net-flow operationalization 2026-05-19); the subset with non-zero subsidy under either metric is N=23 (post-2026-05-17 cycle 6 expansion from N=22; live recompute against sibling regression CSV verified). Pearson $r=0.59$ with Livepeer included and $r=0.01$ with Livepeer excluded (post-2026-05-17 cycle 6 cascade per commit `46d79ee0` L2732-2734 cascade; r inclusive was 0.42 + r excluding was 0.008 in R2-2026-05-17 intermediate state). This broader combined-sample sensitivity replicates the main Section 3.7 finding qualitatively: subsidy correlates with HHI only through Livepeer's 88.5x subsidy outlier.

Manuscript changes: Table 5 HHI-Gini row N updated to 40 with all-protocol Gini coverage; Table 5 TT-expanded subsidy row N corrected from 22 to 19 in R2-2026-05-17 cycle then further expanded to N=20 with $r=0.12$ in the post-2026-05-17 cycle 6 finalization workstream; Section 3.7 paragraph updated with both N=20 TT-direct and N=26 combined-sample sensitivity (R2-2026-05-17 intermediate state: N=19 + N=25; cycle 12 net-flow operationalization 2026-05-19 expanded both); Table 4 expanded from 37 to 40 protocols with Hivemapper / io.net / Aethir as full rows; supplementary exhibit phase0_data_collection_results_2026-05-17.md documents Phase 0 methodology; supplementary exhibit holders_ATH_2026-05-17.csv ships with R2 submission.

Issue 4: Figure-vs-text statistical drift (Figures 4, 5, +)

Reviewer comment (verbatim, key points)

Figure 4 visually reports Mann-Whitney $p = 0.016$, Cohen's $d = 0.96$, DeFi mean = 0.043, and DePIN mean = 0.090. However, the caption and revised text report $p = 0.014$, $d = 1.03$, DeFi mean = 0.041, and DePIN mean = 0.091. Figure 5 has title reporting $r = 0.19$, $p = 0.25$, while the caption and abstract report $r = 0.18$, $p = 0.28$. The same goes for other figures, too. My general suggestion is to reproduce all the tables and figures, and make the text reflect the current data.

Response

The reviewer is correct on the specific Figures 4 and 5 drift and on the general methodology suggestion. The R2 cycle adopts a “reproduce all tables and figures from canonical regression dataset” discipline. The figure regeneration cycle is in flight: the R2 cycle has fully aligned the body text and table values with the canonical regression dataset; the figure PNG regeneration is the remaining Phase 4 deliverable and will be applied to the R2 submission package as a parallel workstream to the manuscript surgery. The regenerated figures will produce visual statistics that match caption and body-text values exactly across all six figures (3, 4, 5, 6, 7, 8). The replication GitHub repository linked in the submission cover contains the regeneration scripts (Python with matplotlib).

Methodology adopted for R2 onward. Per the reviewer's general suggestion and to prevent recurrence, the R2 cycle adopts a single-canonical-CSV-of-record discipline: all manuscript surfaces with embedded statistics regenerate from one canonical regression dataset CSV (shipped as Supplementary File S5). The Section 2.10.3 methodology note documents this discipline, including a snapshot-date discipline acknowledging that voting-power distributions can drift over multi-month windows (the May 2026 Tally pull confirmed this empirically for Compound).

Manuscript changes: Section 2.10.3 figure-regeneration discipline note added; canonical regression dataset CSV updated with Phase 0 additions (Hivemapper, io.net, Aethir Gini values; TT-expanded subsidy N reconciliation); regeneration scripts to be committed in the R2 submission package per the figure-regeneration Phase 4 workstream.

R2 substantive enhancements beyond Reviewer 1 issues

The R2 cycle also includes substantive philosophical-methodological strengthening that goes beyond the four issues Reviewer 1 raised. These are summarized below for transparency.

Predominant-amplification thesis with two structural exceptions (Section 3.5). The Issue 1 resolution plus the post-2026-05-17 deep-PCA-audit cycle (commit 3a8c3065; 2026-05-19; Table 7 sample expanded from N=8 to N=10 via GMX and ENS Tally additions) plus the cycle-13 voting-HHI methodology refresh (commit 99036422; 2026-05-21; Snapshot all-proposals + Tally top-1000) produces a new headline finding: delegation amplifies governance concentration in eight of ten protocols in the cycle-13-refreshed Table 7 sample (amplifier range 2.7x to 9.9x; mean 5.4x; ENS at 0.45x and GMX at 0.87x as structural exceptions). Section 3.5.1 documents the ENS case as a mature-delegate-program counterexample (institutional-design-tractable existence-proof); Section 3.5.1.1 NEW documents the GMX case as a methodology-sensitive counterexample where the broader Tally top-1000 sample reveals dispersion that the top-100 sample obscured (sampling-depth artifact). Three sequential cycles refined the headline: R2-2026-05-17 “universal across 8 protocols 1.9x-6.8x mean 4.1x”; R2-2026-05-19 deep-PCA-audit “near-universal across 9 of 10, 1.2x-11.4x mean 5.3x, ENS exception”; R2-2026-05-21 voting-HHI methodology refresh “predominant across 8 of 10, 2.7x-9.9x mean 5.4x, ENS + GMX exceptions”. Section 3.5 is rewritten to reflect the predominant framing; Section 3.5.1 + 3.5.1.1 added for the two structural-exception cases; Section 3.5.2 documents the ve-token class (Curve veCRV 15x; Balancer veBAL 21x) as extreme amplification (separate methodology not affected by cycle-13 refresh); Section 3.7 adds a PCA-symmetric robustness check; NEW Section 2.10.3 voting-HHI methodology refresh subsection documents the all-proposals + top-1000 methodology.

Philosophical-credibility two-layer framing (Section 2.10.1; Table 1 caption). The R1 Table 1 caption glossed “Pettit Contestation (freedom from arbitrary power)” but “freedom from arbitrary power” is the textbook definition of non-domination, not contestation; the gloss-vs-label mismatch was an internal inconsistency. The R2 cycle resolves this by adopting an explicit two-layer framing: each lens has an operational mechanism (what we score in Table 1) and a philosophical goal (the normative warrant the tradition supplies). The scoring rubric measures the mechanism; the philosophical tradition supplies the warrant. The Pettit row keeps “Contestation” as the label (matching the scoring criteria: appeals + veto rights) and clarifies that non-domination is the goal-state contestability is designed to produce. The two-layer structure is documented in Section 2.10.1 and applied consistently across all five lenses; the Synergy Index reference paragraph (Section 2.10.6) is updated to reflect this.

Inter-lens relationships paragraph (Section 2.9.6). New paragraph documenting that the five lenses are mutually reinforcing rather than orthogonal: publicity (Kant) enables contestability (Pettit); polycentricity (Ostrom) enables knowledge-use (Hayek); Difference Principle (Rawls) constrains contestation outcomes; non-domination (Pettit) requires publicity (Kant) per Pettit 1997 §4.3. The Synergy Index’s arithmetic-mean aggregation reflects this mutually-reinforcing structure; an alternative weakest-link aggregation (geometric mean) is identified as a structurally-defensible alternative.

Systematic empirical-philosophical mapping (Section 4.1). For each major empirical finding, we now state philosophical implications across applicable lenses. The universal delegation

amplification finding (Section 3.5) registers across all five lenses simultaneously: it thins Kantian publicity, attenuates Pettit contestability in practice, undermines Rawlsian fairness, inverts Ostromian polycentricity, and silences Hayekian knowledge-use.

Comparative methodology against existing DAO assessment frameworks (Section 4.5). New paragraph positioning the five-lens framework against Aragon DAO Health Index, Snapshot voting analytics, Boardroom, and DAOstar standards along three dimensions: scope (philosophical-tradition anchor vs operational-only), aggregation (Synergy Index vs single-dimension), and comparability (uniform cross-protocol methodology vs per-protocol dashboards).

Falsifiable forward claims + pre-registration intent (Section 4.9). Two falsifiable predictions: (i) DePIN protocols entering the sample in 2026-2027 will exhibit higher holding HHI than DeFi protocols entering in the same period (conditional on launch year and insider allocation); (ii) any protocol introducing a delegate-program after R2 will produce voting HHI > holding HHI within 12 months of program launch. Future panel-data and event-study work testing these predictions will be pre-registered.

Data-availability and reproducibility statement (Section 4.9). Explicit statement that the 40-protocol governance concentration dataset, exclusions log, regression dataset, Phase 0 supplementary data (Hivemapper / io.net / Aethir holder distributions; veCRV cumulative deposits; Theil/Atkinson supplementary metrics) are all available in the linked replication repository.

Methodology citations (Section 3.7). Added Theil (1967), Atkinson (1970), Hirschman (1964) as historical sources for the inequality indices used in the post-exclusion robustness check. Added Shapley & Shubik (1954) and Banzhaf (1965) as power-index references contextualizing the HHI choice. Added Holmstrom (1979) as principal-agent foundation for the Section 3.5 delegate-program analysis.

Multiple-comparisons correction note (Section 3.7). Table 5 reports 14 separate tests; we now report that applying Benjamini-Hochberg false-discovery-rate correction at $q=0.05$ leaves three of four significant findings (the subsidy-with-Livepeer result is fragile under both multiple-comparisons correction and the Livepeer-outlier sensitivity already noted).

Operationalization-dependency caveat (Section 4.8). New paragraph acknowledging that the framework's empirical conclusions depend on operationalization choices; alternative operationalizations of each lens could yield different lens scores. The HHI cross-protocol rank ordering is robust to choice of concentration metric (Theil and Atkinson correlations in Section 3.7); the framework's lens-level rank ordering may be more sensitive to operationalization choices, an open empirical question for inter-rater reliability and operationalization-robustness analysis.

Citations added (21 total across R2 cycle). Theil (1967), Atkinson (1970), Hirschman (1964), Shapley & Shubik (1954), Banzhaf (1968), Holmstrom (1979), Hardin (1968), Olson (1965; existing citation made more central), Skinner (1998), Cohen (1989), Habermas (1996), Berlin (1958), Beitz (1979), Pogge (2002), Pettit (2012), Rawls (2001), Brennan & Buchanan (1985), North (1990), Atzori (2017), DuPont (2014), Schneider (2024), Tirole (2001). [Update 2026-05-22: Schneider citation year corrected from 2019 to 2024 per CRITICAL year-fix in commit

31230377; Banzhaf year corrected from 1965 to 1968 per Pattern 27 Mode B verification of canonical Banzhaf citation; updated reference is “Schneider, N. (2024). Governable spaces: Democratic design for online life. University of California Press (Luminos Open Access). <https://doi.org/10.1525/luminos.181>”.]

Post-2026-05-17 cycle additions (2026-05-22)

Four post-2026-05-17 cycles further refined the manuscript without altering the Reviewer 1 R2-flagged scope. See the accompanying cover letter 2026-05-17_R2_cover_letter.md for the cycle 5 (deep-PCA-audit), cycle 6 (LFU 2300 absorption), cycle 13 (voting-HHI methodology refresh), cycle 14 (title rename + COI correction + word-cap trim cycle), and cycle 15 (multi-source PCA-symmetric extension + 8-pass deep-read audit + tracked-changes readability fix) narratives.

The 2026-05-22 cycle 14 added 6 new supplementary files (S10, S11, S12, S15, S16, S17) capturing computational detail and per-protocol narrative depth, enabling a targeted main-paper trim from 17,702 to ~16,321 words while preserving all load-bearing inferential defenses and the framework-empirical bridges elevated by the DEC-190 title rename. The new supplementary files are enumerated in the cover letter Submission package contents section.

The 2026-05-22 cycle 14 also corrected the Conflict of Interest statement in Section 2.10.8 to reflect that the author’s tenure at Borderless Capital ended February 2026 (the as-R2-submitted statement carried stale present-tense framing). The corrected COI is documented inline in Section 2.10.8 of the manuscript and at the close of the accompanying cover letter.

The 2026-05-22 cycle 14 renamed the manuscript title from “Governance Concentration Beyond Token Allocation: An Institutional Design Analysis of DePIN and DeFi” to “Tokenomics as Applied Political Philosophy: Governance Concentration Beyond Token Allocation” per DEC-190 to better reflect the paper’s stated core thesis (Section 1.2). The empirical content, methodology, and conclusions are unchanged from the R2 resubmission shipped 2026-05-12.

Cycle 15 substantive additions (2026-05-22; multi-source PCA-symmetric extension + max-effort deep-read audit + tracked-changes readability fix)

Subsequent to the cycle 14 trim, a four-part cycle 15 landed 2026-05-22 substantively strengthening reviewer-defensibility while preserving all R2 headline findings. Cycle 15 does NOT alter the Reviewer 1 R2-flagged scope; the new findings extend the §3.5 / §3.7 robustness checks per the reviewer’s general suggestion that table-vs-text drift be eliminated via comprehensive reproduce-and-sync, and add three new findings (F-B2-13 / F-B2-14 / F-B2-15) per a “max defensibility” extension. The four parts:

Cycle 15.1: Multi-source PCA-symmetric extension (commit 1751fd5c workflow + d4366a1 replication). The §3.7 PCA-symmetric robustness check extended across three independent governance-data sources beyond the original Tally-only specification: (a) Tally all-delegates pull for 4 of 10 protocols at full depth (AAVE N=150,911; COMP N=18,627; UNI

N=48,707; ARB N=437,453; all four sub-0.005 HHI shift vs cycle-13 top-1000 baseline, empirically validating the top-1000 methodology floor); (b) Snapshot 13-protocol voter-pool extension with cross-source label / Blockscout MCP / cross-protocol-overlap analysis (Tracks A through D); (c) Solana on-chain governance program parsing for JUP / HNT / DRIFT (JUP via Dune `jupiter_solana.govern_call_setvote` per-signer max-weight aggregation; HNT and DRIFT via Helius RPC parsing of SPL Governance VoteRecordV2 accounts).

A new sibling canonical store `data/processed/exclusions_log_signer.csv` registers signer-side functional-PCA voters (Compound Foundation `0xb06df4dd...`; Balancer DAO multisig `0xad9992f3...`; Kevin Owocki / Gitcoin Maintainer `0x00de4b13...`; WXM Hedgey-vested insider `0x96d8901a...`; 4 confirmed entries after the DIMO Foundation candidate was REFUTED via Polygon Blockscout MCP funding-source verification per a hypothesis-revision pattern documented in EC-2026-05-22-B2-WXM-Snapshot-Top1-Identification + sister EC entries).

A methodologically substantive finding emerging from the Snapshot signer-side PCA-symmetric recompute: PCA-symmetric exclusion is NOT monotonic in the concentration-reduction direction. For three of four Snapshot-sourced protocols with confirmed signer-side functional PCAs, exclusion produces the expected concentration reduction: COMP HHI shifts from 0.0889 to 0.0468; BAL from 0.3045 to 0.1499; WXM from 0.5561 to 0.1184. For the remaining protocol, signer-side PCA exclusion INCREASES voting HHI: GTC shifts from 0.1792 to 0.1948 (smaller-denominator effect when excluded PCA holds dominant share). The methodological implication: PCA-symmetric exclusion is a substantive direction-of-effect test rather than a mechanical concentration-reduction adjustment.

Three new EC entries codified in Cycle 15.1: EC-2026-05-22-B2-WXM-Snapshot-Top1-Identification (hypothesis-revision audit trail; original WXM Foundation operational-signer candidate refined to Hedgey-vested insider allocation recipient via Blockscout MCP funding-source verification); EC-2026-05-22-B2-HNT-Vote-Count-Proxy-vs-Lockup-Weight-Drift (methodology-floor drift subclass; 5-7.5x shift between vote-count proxy 0.00525 and proper VSR lockup-weight 0.0261-0.0394; HALT-1.3 triggered as dispatch designed); EC-2026-05-22-B2-DRIFT-Realms-Mint-Citation-Correction (surface-extraction-vs-on-chain-truth drift; new sub-class; DriftCq mint cited from Realms DOM does not exist on-chain; true mint DriFtupJ verified via Helius `getAccountInfo`).

Cycle 15.2: Table 7 N=10 to N=13 expansion + F-B2-9 reframe (commit 37635794). Per the multi-source extension, three Solana-native protocols added to Table 7: JUP delegated-vote 0.057x dispersion; HNT VSR lockup-weighted 0.35x-0.53x sensitivity-range dispersion; DRIFT VSR lockup-weighted 1.6x mild amplification. Aave row updated from cycle-13 top-1000 0.062 to all-delegates 0.058 (4.4x ratio; methodology-floor empirical-validation tolerance under 0.005). F-B2-9 framing reframed from “8 of 10 amplify with 2 structural exceptions” to **“9 of 13 amplify (range 1.6x DRIFT to 9.9x Compound; mean approximately 4.9x) with 4 structural exceptions (ENS 0.45x mature-delegate-program; GMX 0.87x methodology-sensitive; HNT 0.35x-0.53x Solana VSR-lockup-weighted; JUP 0.057x most-extreme dispersion outlier with Solana delegated-vote + broad-airdrop-user-base mechanism)”**. Substantive thesis evolution: JUP at 0.057x is the new most-extreme dispersion outlier (8x more extreme than prior most-extreme ENS at 0.45x), introducing a fundamentally new design pattern producing concentration

order-of-magnitude below the cross-section median. F-B2-9 predominant-amplification thesis PRESERVED at 9-of-13 majority.

Three new PAPER.md subsections added in Cycle 15.2:

- **§3.5.4 Within-Solana governance concentration heterogeneity:** documents the 15x within-Solana spread driven by VSR-vs-delegated weight design as primary axis (HNT + DRIFT VSR cluster at EVM-typical concentration; JUP delegated-vote with broad airdrop user base produces low-concentration outlier). Methodology-acknowledgment that chain context alone (Solana-ness) is insufficient to produce low-concentration governance; chain-tied protocol designs structurally confound clean protocol-level-vs-chain-level decompositions (the original “JUP-specific mechanisms RATHER THAN chain-level mechanisms” framing was internally inconsistent and was refined per EC-2026-05-22-B2-F-B2-14-Chain-vs-Protocol-Design-Dichotomy-Overclaim).
- **§3.5.4.1 Drift Foundation proposal-authorship and vote-weight decoupling:** documents Drift Protocol’s governance pattern with no close analog in EVM-protocol cross-section. A single Drift Foundation submitter address (4CL25...wPiif) proposed 11 of 13 DRIFT Improvement Proposals over the 12-month window while holding only 0.18 percent of total DRIFT vote weight; proposal-authorship and vote-weight are structurally decoupled.
- **§3.5.5 Cross-protocol institutional-governance-investor pattern:** PGov + Tane Governance + Arana Digital collectively exert approximately 17 percent combined Snapshot voting power across 6-7 DAOs each; Tally-side cross-protocol overlap analysis confirms the pattern at 8.11 percent combined summed share across ARB+COMP+UNI plus 7 additional institutional governance firms (Blockworks Advisory; Wintermute Governance; L2BEAT; Humpy; Gauntlet; GFX Labs; olimpio). Candidate Fifth Concentration Axis alongside the four already in B2 scope (holder + voter + delegate-class + burner concentration).

Figure 6 PNG regenerated with N=13 data via the canonical `tools/content-build/b2_figure6_regen.py` script. Figure 3 caption refreshed to current canonical (133 PCAs / 38 protocols post-cycle-4 audit; Aethir 0.095 / WeatherXM 0.148 / io.net 0.125 top DePIN cluster after IoTeX dropped from 0.189 to 0.081). 12+ prose surfaces cascaded with N=13 framing across abstract Finding 4, §1.4 fourth finding, §3.5 narrative, §3.7 amplification sub-text, §4.1 inter-lens implications, §4.6 summary of findings, §4.7 contributions, §5/6 governance design implications.

Cycle 15.3: Max-effort 8-pass deep-read stale-anchor audit (approximately 80 anchors cleared). A systematic 8-pass audit cleared cumulative stale anchors arising from the multi-cycle revision arc: Pass 1 wording-variation residuals (13 surfaces); Pass 2 Table 7 source-attribution completeness (1 footnote refresh); Pass 3 protocol-specific HHI cycle drift (7 surfaces; Curve 0.017 to 0.014 raw post-Convex-exclusion; Jupiter max-DeFi HHI 0.117 to 0.096; Hyperliquid minimum 0.004 to 0.005; Figure 3 caption pre-fix IoTeX 0.189 cluster reference); Pass 4 sequential-read in-context inconsistencies (13 surfaces; abstract Finding 4 enumeration; §1.4 amplifier list missing DRIFT; §3.3 DePIN mean 0.077 pre-IoTeX-fix to 0.071 post-fix; WeatherXM voting 0.486 to 0.556; “five DeFi protocols” to “seven”; Arbitrum amplification 4.4x typo to 3.1x; ENS top-1 19.9% pre-Cold-Wallet-fix to 16.8%); Pass 5 additional cross-

checks (5 surfaces; Hivemapper 0.017 to 0.018; Morpheus 0.031 to 0.046; AAVE delegation HHI 0.076 to 0.058; Figure 8 $r=0.00$ to $r=-0.10$); Pass 6 systematic Python-based Table 4 row vs sibling-clone replication CSV canonical (4 substantive row corrections including Aethir HHI 0.0868 to 0.0948 with updated Top-1 19.7% / Top-10 67.6% reflecting cycle-4 0xfc78 Safe multisig exclusion that Table 4 caption already documented; LayerZero / Arbitrum / Optimism row refreshes); Pass 7 abstract-vs-Table 5 internal consistency (3 surfaces; Table 5 insider count fraction row $\rho=0.44$ to 0.48 / $p=0.005$ to 0.003 / $N=39$ to 37 to match abstract canonical); Pass 8 Table 4 N column convention per author directive (33 row updates: N now represents pre-exclusion sample size = 1,000 + number of PCAs; e.g., Optimism $N = 1,007$ reflects top-1,007 pre-exclusion sample with 7 PCAs excluded to yield 1,000 non-PCA holders for HHI computation). Plus Pass 8 extension: value-vs-convention methodology footnote acknowledging HHI / Top-N values reflect top-1000 pre-exclusion computation while N column reflects (1000+PCAs) convention; strict-convention recomputation would yield < 0.0001 HHI shift / < 0.1 pp Top-N shift (immaterial for cross-protocol comparisons).

Substantive corrections caught in the audit (highlighted because they touch the manuscript's quantitative core): F-B2-14 (JUP outlier) "JUP-specific mechanisms rather than chain-level mechanisms explain the outlier" framing was internally inconsistent with the "low effective transaction cost" design implication (low tx cost IS a chain-level feature); refined to "JUP's design choices are protocol-design-level decisions ENABLED BY Solana's low-cost voting environment; whether they would replicate at scale on a higher-fee chain is untested because no Ethereum counterfactual exists" with a new EC-2026-05-22-B2-F-B2-14-Chain-vs-Protocol-Design-Dichotomy-Overclaim documenting the methodology lesson. F-B2-15 (within-Solana heterogeneity) DRIFT mis-classification corrected (was "delegated-vote with smaller professional user base"; correct is "VSR lockup-weighted vote design analogous to HNT" per DRIFT memo line 125); mechanism story reframed around VSR-vs-delegated as primary within-Solana driver per DRIFT memo line 130; sister EC-2026-05-22-B2-F-B2-15-DRIFT-VSR-Classification-And-Within-Solana-Mechanism-Refinement filed. The audit-pass discipline established by these corrections (verify just-shipped findings via Pattern 27 Mode A within-cycle) is itself an internal methodology contribution that addresses adjacent concerns to Reviewer 1's general "reproduce and sync" recommendation.

Cycle 15.4: Tracked-changes readability fix (commit 859ebb64). The previous tracked-changes generation wrapped EVERY text run in `w:ins` markers (8,462 markers) which made the tracked-changes view unreadable for reviewers: every paragraph appeared edited, even those merely split from R1 mega-paragraphs during the jargon-refinement / accessibility cycles. A new diff-aware generator (`tools/content-build/regenerate_tracked_changes_v3.py`) performs paragraph-level token-set similarity matching between R1 baseline and R2 clean DOCX, wrapping `w:r` elements ONLY in paragraphs with substantive content changes (Jaccard ≥ 0.50 OR coverage ≥ 0.70 to classify as unchanged; otherwise wrap). Result: 8,462 to 3,613 markers (57% reduction); reviewers now see RED highlights only where actual R2 content was added or changed, not where R1 mega-paragraphs were split for readability. The 192 marked paragraphs capture the substantive R2 changes (new §3.5.4 / 3.5.4.1 / 3.5.5 subsections; §3.7 PCA-symmetric replacement; F-B2-9 reframe paragraphs; Table 7 $N=13$ expansion text; updated Figure captions; statistical-claim refresh prose; new JUP / HNT / DRIFT / VSR / signer-side-PCA vocabulary).

Cycle 15 method-of-record discipline: single-canonical-regression-dataset-of-record discipline (Section 2.10.3) maintained throughout the multi-source extension; sibling clone `regression_data_april2026.csv` remains the holding-HHI canonical; new `exclusions_log_signer.csv` registered as sister canonical for signer-side voting-symmetric exclusion. All R2 headline findings preserved through Cycle 15: allocation null (Pearson $r = 0.05$, $p = 0.76$, $N = 37$); DePIN-DeFi sector contrast (Mann-Whitney $p = 0.020$, Cohen's $d = 0.94$; LOO 30/30); subsidy disconnect (Livepeer outlier; $r = 0.59$ inclusive / $r = 0.03$ exclusive); predominant delegation amplification (now 9 of 13 amplify; 4 structural exceptions: ENS + GMX + HNT + JUP); cross-protocol institutional concentration emerging as candidate Fifth Concentration Axis (per Section 3.5.5).

Cycle 16 substantive additions (2026-05-22; revision-artifact cleanup + Frontiers 15-figure/table-limit compliance)

A final manuscript-polish cycle landed 2026-05-22 immediately prior to this submission, addressing two reviewer-defensibility concerns:

Cycle 16.1: Revision-artifact cleanup (15+ surfaces). The manuscript body previously contained revision-cycle annotations carried forward from the multi-cycle revision process (Cycles 5 through 15). Final-submission manuscript text should NOT contain revision-cycle references (cycle-13, cycle-14, cycle-15, pre-refresh, post-refresh, post-cycle-4, etc.) or audit-trail provenance for intermediate sampling depths; revision-cycle context belongs in this point-by-point response document + the accompanying cover letter + the linked supplementary materials, not in the manuscript body. Reviewers reading the manuscript should encounter current-state framing only.

Surfaces cleaned in Cycle 16.1: - §2.10.3 voting-HHI methodology refresh subsection: pre-refresh / post-refresh narrative replaced with canonical current-state framing (“Voting-HHI sampling methodology”); per-protocol sample-size comparison table replaced with current-state “Table 8. Per-protocol active-voter sample size and methodology source” (single N column; per-source-methodology notes; 13 protocols including the three Solana-native additions). - Figure 3 caption: removed “69 PCAs / 20 protocols preserved as cycle-13 pre-deep-audit baseline” revision-history annotation. - Table 4 caption: removed “(post-cycle-4 audit adding 0xfc78 Safe multisig)” parenthetical revision-history annotation from the Aethir note. - Lido amplification note (§3.5): removed “(post-refresh value per the 2026-05-21 voting-HHI methodology revision; pre-refresh 11.4x preserved as audit trail in the program canonical-state record)” parenthetical; now reads simply “Lido’s 6.3x amplification” with substantive discussion preserved. - §3.5.1.1 GMX header + body: removed “(post-refresh)” header annotation; removed “commit chain pre-2026-05-21” + “The 2026-05-21 voting-HHI methodology refresh” framing from body prose; replaced with current-state framing. - §3.5 DeFi enumeration: removed “at cycle-13 refresh” annotations from Compound and Arbitrum amplification references. - §3.5.4 prose: removed “in the R2 revision cycle” qualifier from the Solana-protocols introduction. - §3.5.4 within-Solana spread: “across the cycle-13 voting-HHI cross-section” updated to “across the EVM voting-HHI cross-section”. - Figure 6 caption: removed “The figure visualizes the cycle-13 ten-protocol cross-section; the multi-source extension cascade (2026-05-22) adds three Solana-native protocols...” revision-history narrative; condensed to current-state $N=13$ framing. - Table 7 † footnote: removed “(cycle-13 refresh yields $n = 138$ and $n = 5,466$ unique voters in the 12-

month rolling-window measurement period respectively per the 2026-05-21 methodology refresh; March 2026 baseline $n = 114$ and $n = 5,241$ preserved as audit-trail per Section 2.10.3) multi-line audit-trail revision history; replaced with current-state framing. - Table 7 § footnote: removed “(N = 150,911 at 2026-05-22 multi-source extension cascade; supersedes cycle-13 top-1000 value 0.062 with HHI shift -0.004 within the top-1000 methodology-floor empirical-validation tolerance)” + “vs cycle-13 top-1000 baseline, confirming the methodology floor” revision-history phrases. - §3.2 Morpheus reference: removed “via sibling clone regression dataset 2026-05-22” revision-artifact source annotation. - §3.7 PCA-symmetric paragraph: removed “cycle-13 top-1000 sample baseline” annotations (2 places). - §3.7 Shapley-Banzhaf: removed “the original Shapley computation at top-100 sample yielded 0.076 pre-2026-05-21 methodology refresh” revision-history parenthetical. - §2.10.3 Snapshot date discipline paragraph: removed multi-paragraph revision-history narrative (“Replication Tally and Snapshot API queries in May 2026 indicated... compared with the 100-delegate sample available in March 2026. The Snapshot voter pool for Compound grew from 114 unique voters in March 2026 to 138 by May 2026...”) and condensed to a single drift-acknowledgment sentence. - F-B2-9 internal identifier references (2 surfaces in §3.5.5 + §3.7): replaced with “predominant-amplification finding (Section 3.5)” current-state framing. - Dated supplementary script file paths (7 surfaces): “b2/paper/supplements/synergy_index_full_sample_2026-05-19.py” / “power_indices_2026-05-19.py” / “subsidy_multivariate_2026-05-19.py” / “profitability_retention_2026-05-19.py” / “net_flow_burn_data_2026-05-19.csv” / “profitability_retention_summary_2026-05-19.csv” replaced with S-supplement prose references (e.g., “supplementary repository power-indices reproducibility script”). - Dated price-snapshot references: “as of 2026-05-19” replaced with “as of the May 2026 data-collection snapshot”; “at the 2026-05-19 POL/HNT price” replaced with “at the May 2026 POL/HNT price”. - §4.8 audit-cycle narrative: “The 2026-05-19 universal verification audit cycle surfaced three per-protocol PCA-exclusion sensitivity-check items...” multi-sentence revision-history paragraph condensed to current-state framing “Per-protocol PCA-exclusion sensitivity is documented for three protocols at the upper end of the DePIN concentration distribution” with audit-trail detail preserved via the existing Supplementary File S16 reference.

Preserved legitimate methodology context (NOT removed): - “March 2026 snapshot” in Table 4 / Figure 3 captions (data-collection date specification) - “12-month rolling window 2025-05-22 to 2026-05-22” in Snapshot methodology - 1 pre-existing em-dash (source-fidelity exception)

Post-cleanup verification: zero instances of cycle-13 / cycle-14 / cycle-15 / pre-refresh / post-refresh / post-cycle-4 / preserved-as-audit-trail / supersedes / multi-source-extension-cascade / pre-2026-05-21 / in-the-R2-revision-cycle / F-B2-N / EC-2026-NN / DEC-NNN patterns in the manuscript body.

Cycle 16.2: Frontiers 15-figure/table-limit compliance. Frontiers Original Research and Review Articles specifies a maximum of 15 figures/tables. The pre-Cycle-16 manuscript carried 9 figures + 9 tables = 18 (over the maximum). Three items removed to bring the count to exactly 15 (at limit):

- **Figure 8** (HHI vs voter participation; $r = -0.10$ null; ancillary finding). The participation-vs-concentration null is briefly discussed in §3.7 text; the figure showed an essentially flat slope across 13 protocols. Per-protocol participation data preserved in Supplementary File S5 pipeline specification. Removal preserves the discussion: “Governance

concentration shows no association with voter participation rates ($r = -0.10$, $N = 13$ across 13 protocols with Snapshot or Governor governance data; per-protocol participation data in Supplementary File S5”).

- **Figure 9** (LOO forest plot for the DePIN-vs-DeFi sector contrast). Visualizes the Cohen’s d range 0.84 to 1.10 across 30 leave-one-out iterations + bootstrap 95% CI [0.32, 1.68] for the headline d . The robustness discussion is preserved verbatim in §3.7: per-iteration LOO data and bootstrap 95% CI available in Supplementary File S5.
- **Table 9** (PCA-classification robustness; 5-spec exclusion sensitivity). Already documented in [Supplementary File S10](#) with full per-specification narrative + alternative-classification rationale. Table 9 inline content condensed to a prose sentence: “Under progressively stricter exclusion specifications (Specs A through E in Supplementary File S10), Spec A (canonical 5-class typology) yields Mann-Whitney $p = 0.029$ and Cohen’s $d = 0.94$; Spec B (drop Class 5 CEX custody) yields $p = 0.039$, $d = 0.84$; Specs C through E yield $p > 0.10$ with Cohen’s d in the 0.17 to 0.60 range.”

Removability criteria satisfied for all three: (i) ancillary (not headline) finding; (ii) prose-vs-visual redundant (full content reproducible in 1-2 sentences); (iii) existing supplementary file alternative (S5 covers per-protocol participation + LOO iteration data; S10 covers PCA-classification robustness directly); (iv) low reviewer-defensibility cost (no headline finding lost).

Post-Cycle-16 manuscript figure + table inventory:

Tables (8): Table 1 (Scoring rubric); Table 2 (Illustrative rubric scores; 3-protocol worked example); Table 3 (Claims boundary); Table 4 (Governance concentration cross-section; 40 protocols); Table 5 (Bivariate associations); Table 6 (OLS regression); Table 7 (Holding vs voting concentration; 13 protocols); Table 8 (Per-protocol active-voter sample size + methodology).

Figures (7): Figure 1 (Conceptual model: philosophy to mechanisms to outcomes); Figure 2 (Normative framework architecture: 5 lenses to rubric to hypotheses); Figure 3 (HHI bar chart across 40 governance tokens); Figure 4 (DePIN vs DeFi sector boxplot); Figure 5 (Insider allocation scatter; lead-finding null visualization); Figure 6 (Delegation amplification: holding vs voting HHI; $N=13$); Figure 7 (Subsidy ratio vs HHI scatter; Livepeer outlier).

Total: 15 figures + tables (at Frontiers limit).

All R2 headline findings preserved through Cycle 16: allocation null (Figure 5; Table 5); DePIN-DeFi sector contrast (Figure 4; Table 6); subsidy disconnect (Figure 7); predominant delegation amplification (Figure 6; Table 7; Table 8 sample-size methodology support); cross-section concentration (Figure 3; Table 4); framework operationalization (Figures 1 + 2; Tables 1 + 2); methodology disclosure (Table 3).

Acknowledgments

We thank Reviewer 1 for the comprehensive cross-surface audit that surfaced the drift class, and Reviewer 2 for the endorsement of publication. The R2 manuscript’s predominant-amplification

finding with four structural exceptions (Section 3.5; abstract finding 4) is a direct product of Reviewer 1's R2 framing: applying consistent post-exclusion methodology across Table 4 and Table 7 produces a substantive interpretive change that strengthens the paper's main thesis about institutional design at the operator and router level governing governance outcomes. The cycle-13 voting-HHI methodology refresh (commit 99036422 per EC-2026-05-21-B2-Voting-HHI-Sampling-Methodology-Floor) surfaced the GMX structural-exception finding at expanded Tally top-1000 sampling depth. The cycle-15 multi-source PCA-symmetric extension (commit 1751fd5c per the multi-source dispatch arc 2026-05-22) surfaced JUP as the most-extreme dispersion outlier (0.057x; Solana delegated-vote + broad airdrop user base) and HNT as a sensitivity-range dispersion case (0.35x-0.53x; Solana VSR lockup-weighted) at the apples-to-apples Solana on-chain governance-program-parsing depth, plus introduced cross-protocol institutional concentration as a candidate Fifth Concentration Axis (PGov + Tane + Arana approximately 17 percent combined Snapshot voting power) per §3.5.5. The R2 cycle's philosophical-credibility strengthening (Section 2.10.1 two-layer framing; Section 2.9.6 inter-lens relationships; Section 4.5 comparative methodology) addresses internal coherence questions adjacent to the reviewer's concerns; these enhancements also extend from the cross-surface audit discipline the reviewer's framing recommended. The cycle-15 8-pass deep-read stale-anchor audit (~80 anchors cleared across wording-variation, source-attribution, protocol-specific HHI cycle drift, sequential-read in-context inconsistencies, Table 4 row vs canonical, abstract-vs-Table 5 consistency, and N column convention layers) and the cycle-15 tracked-changes readability fix (8,462 to 3,613 w:ins markers; 57% reduction) operationalize the reviewer's general "reproduce all tables and figures and make the text reflect the current data" recommendation at saturation depth.